

NexSys PCS[®]

Site Administration Guide

For use with device LN PCS-300-US and
software revisions PCS300.1.3.2 and 1.4.1

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Introduction

What Is the Purpose of This Guide?

The *Haemonetics® NexSys PCS® Site Administration Guide* is intended as a guide for plasma center administrators to provide them with the information needed to configure the NexSys PCS device. It includes the following information:

- A description of the device's configurable settings
- Instructions for changing the configurable settings
- Instructions for connecting the device to a network



Note: The center must have a compatible wired or wireless network, an approved server application (such as HaemoCommunicator™ communication gateway), and a compatible software application (such as Blood Establishment Computer Software [BECS]) to utilize connectivity features.

This guide applies to device list number PCS-300-US and software revisions PCS300.1.3.2 and 1.4.1. Use this guide in conjunction with the user manual and training supplied by qualified Haemonetics personnel.

Symbols Found in This Document

The following symbol is used in this guide to emphasize certain details for the user.



Note: Provides useful information regarding a procedure or operating technique.



Attention: *Advises the user against initiating an action or creating a situation that could result in damage to equipment or impair the quality of the blood products; personal injury is unlikely.*

Device Configuration

Overview

Each NexSys PCS device is preconfigured with Haemonetics default settings. Some of these settings are adjustable and control a range of elements – from basic settings, such as time and date, to more complex settings, such as protocol settings and the location of operator validation screens. These settings can be customized at the device level by a center technician or at the corporate level (if equipped) by a center administrator across all centers and devices in a company.

Device Level

At the individual device level, configuration changes can be made by accessing the device's configurable settings via the *Advanced Settings* screen (described below in "[Accessing a Device's Advanced Settings](#)") and entering them manually or scanning a barcode.

Corporate Level

If a center is equipped with the appropriate infrastructure, including a compatible software system (for example, BECS) and an approved server application (for example, HaemoCommunicator), an administrator can update the configurable settings across all connected devices via a network update.

Accessing a Device's Advanced Settings

To access a device's advanced settings:

1. On the *Initial Setup* screen, touch  (**Settings**) in the top right-hand corner of the screen.

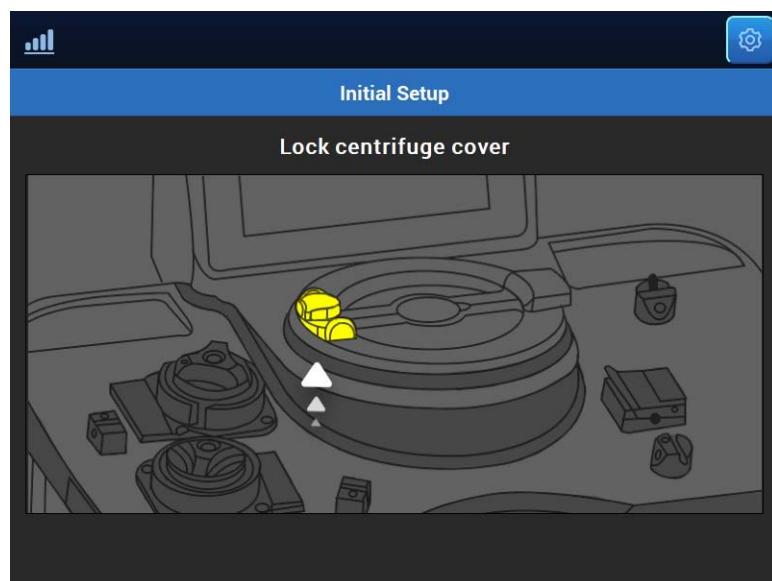


Figure 1, Example of the Initial Setup screen

2. Select **Advanced Settings** from the drop-down menu.

 *Note: You can also access the advanced settings from the Maintenance or Maintenance Required screens. See the Service Manual for more information.*

 *Note: Once you enter the Advanced Settings screen, the only way to exit is to restart the device.*

3. Enter the user ID and password and touch **Verify**.

 *Note: Every device is programmed at the manufacturer with a default user ID and password, which enable you to access the device's advanced settings. Haemonetics recommends that you change these credentials upon initial login. You can do so on the User Settings screen (See ["User Settings" on page 19](#)).*

 *Note: This login prompt does not appear if the Haemonetics Technician USB is inserted.*

4. Touch **Confirm**. The *Advanced Settings* screen appears with a list of available settings and a short description for each.

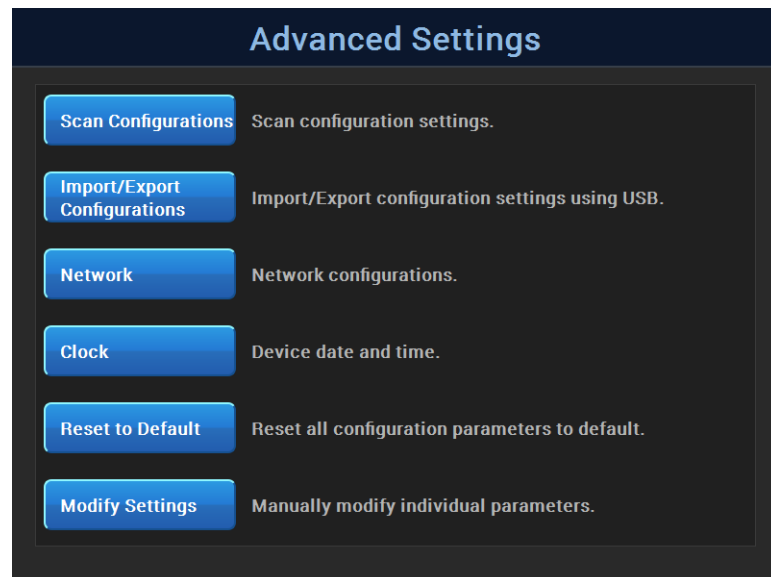


Figure 2, , Advanced Settings screen

 *Note: After making changes to any settings, touch **Save** to save your changes and return to the Advanced Settings screen. You can then select other settings to make additional changes or restart the device to implement the changes.*

Scan Configurations

Touch **Scan Configurations** to access the *Scan Configuration Settings* screen. This screen enables you to change the HaemoOnly configurable settings on your device by using the built-in barcode reader or a Haemonetics approved external barcode reader (For example, the Datalogic® Gryphon™ GD4520-HCK1 external scanner) to scan a QR code. Contact Haemonetics Product Support to obtain a QR code.

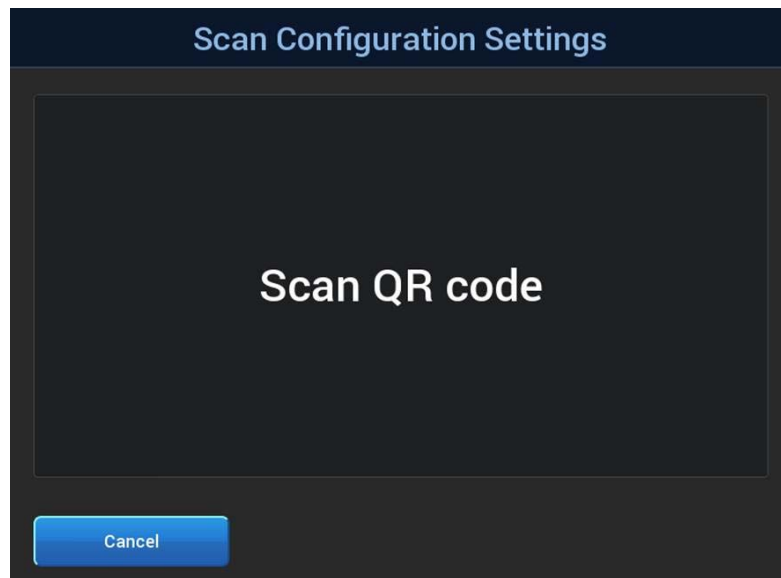


Figure 3, Example of the Scan Configuration Settings screen

To scan configuration settings:

1. Scan the configuration barcode using the barcode reader on the device and touch **Save**.



Note: If the QR code does not scan, try moving the code closer or further away from the barcode reader. The barcode reader light changes to a green dot and an audible tone sounds once the code has been successfully scanned.

2. Review the settings to verify that they are correct.

Import/Export Configurations

Touch **Import/Export Configurations** to choose to configure the device by importing settings from a USB flash drive or to export the current device settings to a USB flash drive. This function copies over all configurations, including network configuration and user authentication settings, except for SN and UDI.



Attention: Do not connect non-approved devices (such as cellular phones, etc.) to the USB ports. Approved devices must comply with IEC/EN 60601-1 or IEC/EN 60950-1.

Network

Touch **Network** to access the *Network Configurations* screen, which provides access to the wireless connection settings for the device that enable it to communicate with an approved server application. For more information on the network configuration settings, see [“Network Configuration Settings” on page 34](#).

Figure 4, Example of the Network Configurations screen

Check the "Connect to BECS" checkbox to enable the device to communicate with a BECS or compatible software system via an approved server application.

To edit the rest of the network configuration settings, select each field and enter the settings manually or scan a network configuration QR code using the barcode reader. When scanning a QR code, the fields autopopulate. Review the settings to verify that they are correct before touching **Save**.



Note: If the QR code does not scan, try moving the code closer or further away from the barcode reader. The barcode reader light changes to a green dot and an audible tone sounds once the code has been successfully scanned.

Once you reboot the device, the changes will take effect. If the device is configured to connect to a BECS or if a server address is configured, the top left-hand corner of the screen displays a network connection indicator:



Connected



Not connected

The screen does not display any symbol if connectivity is not enabled on the device.



Note: In the event of loss of connectivity, the procedure can still continue, but it will follow a disconnected procedure workflow and the screens will be slightly different.

Clock

Touch **Clock** to access the *Date and Time Settings* screen, which enables you to set the date, time, time zone, and time display format for the device.

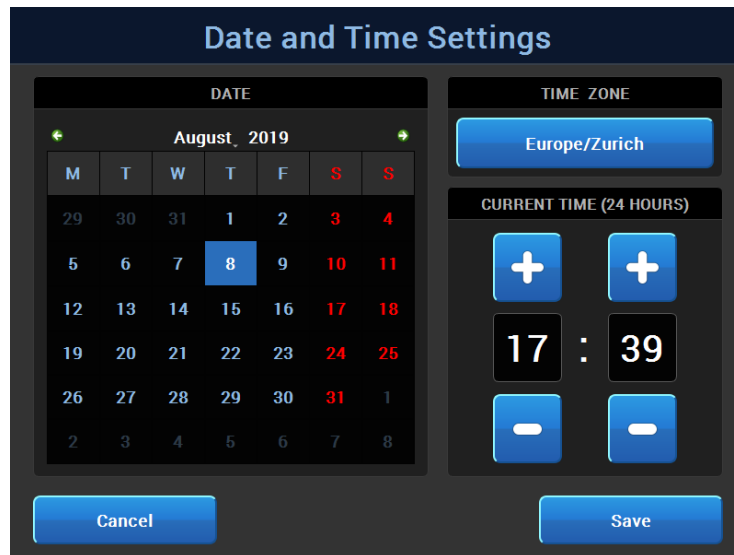


Figure 5, Example of the Date and Time Settings screen

Reset to Default

Touch **Reset to Default** if you want to reset all settings on the device, including the network configuration, to the Haemonetics default setting, and then touch **Confirm**.

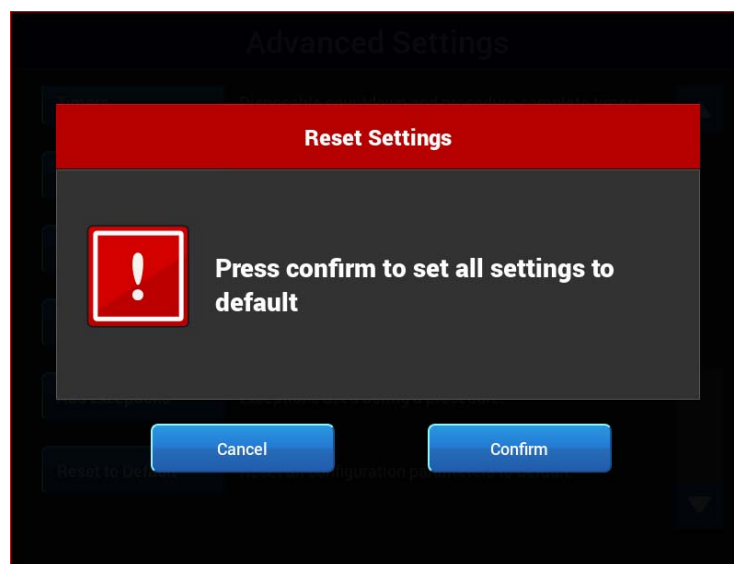


Figure 6, Example of the Reset Settings screen

Modify Settings

Touch **Modify Settings** to view and manually modify procedure settings. These settings are described in detail in "[Modifiable Procedure Settings](#)" beginning on [page 13](#).



Note: The Modify Settings screen is protected by a login prompt that cannot be bypassed with a Haemonetics Technician USB. Modifying any parameters on this screen will affect the Advanced Settings CRC.

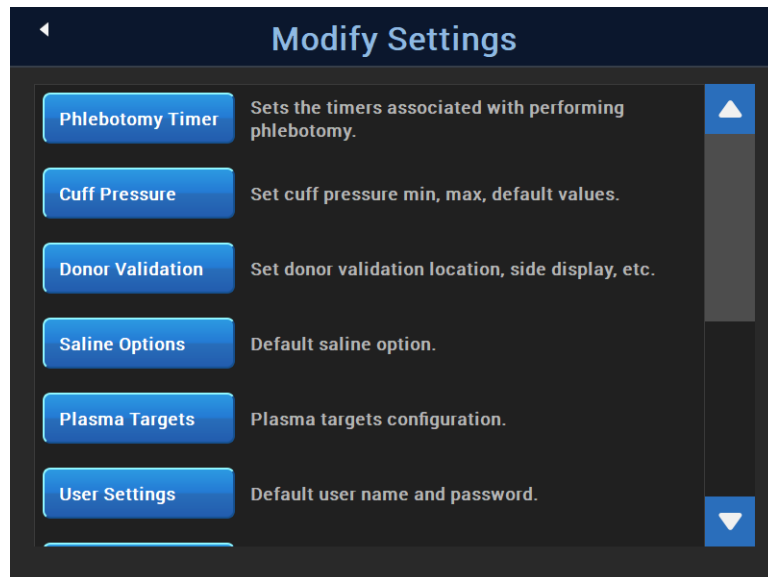


Figure 7, Example of the Modify Settings screen

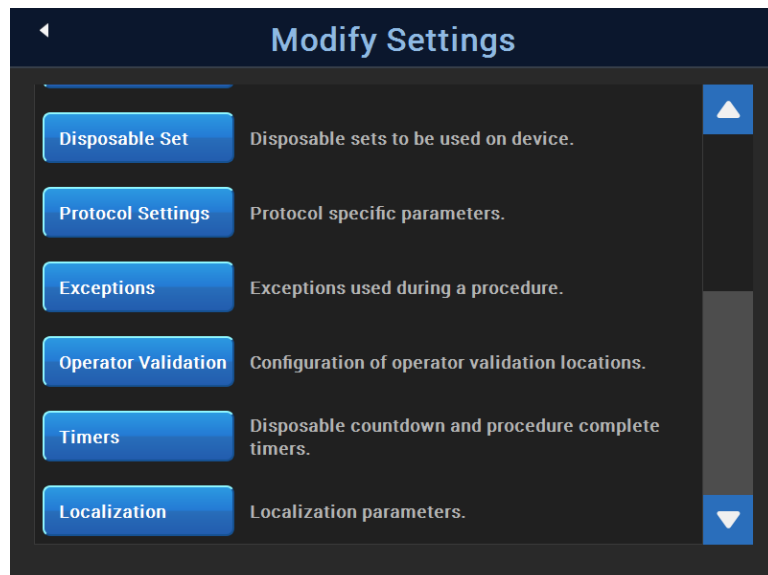


Figure 8, Example of the Modify Settings screen (continued)

Modifiable Procedure Settings

The following procedure settings are accessed via the **Modify Settings** button and control the behavior of the device and the appearance of screens during a procedure.

Phlebotomy Timer

Touch **Phlebotomy Timer** to access the *Phlebotomy Timer* screen. Use the arrows to adjust the limits for the various timers. The timers count down by default. If the "Count-Up Timer" checkbox is selected, all timers except SETUP will count up. If the "Force timer completion before procedure start" checkbox is selected, the **Begin Draw** button will not appear until all timers prior to the venipuncture timer have completed.

If the "Require manual start of consecutive timers" checkbox is selected, each subsequent timer will not start until the user touches **Start**.

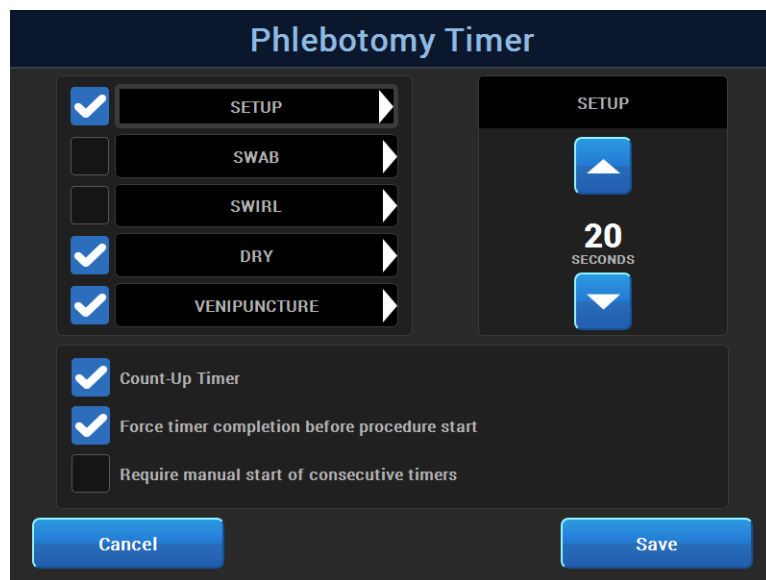
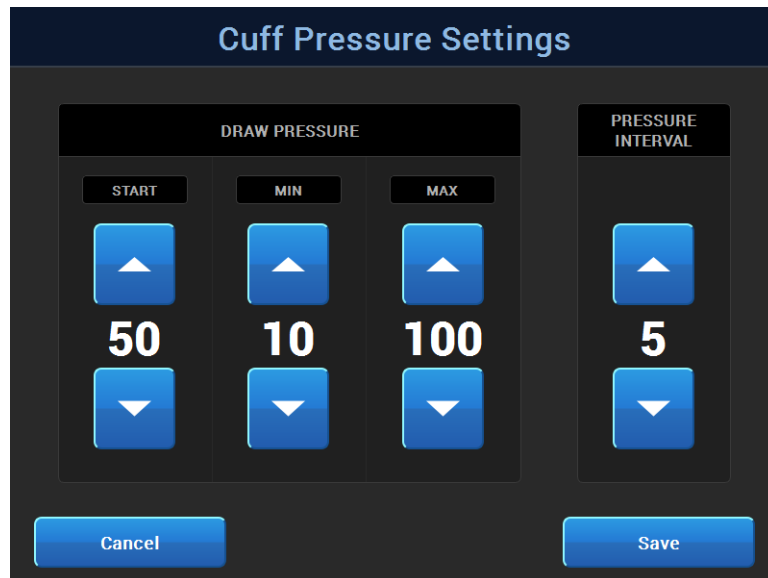


Figure 9, Example of the *Phlebotomy Timer* screen

Cuff Pressure

Touch **Cuff Pressure** to access the pressure cuff settings. The "Draw Pressure" settings control the default start pressure and minimum and maximum values. "Pressure Interval" determines the increment at which the user can adjust the cuff pressure during the procedure.

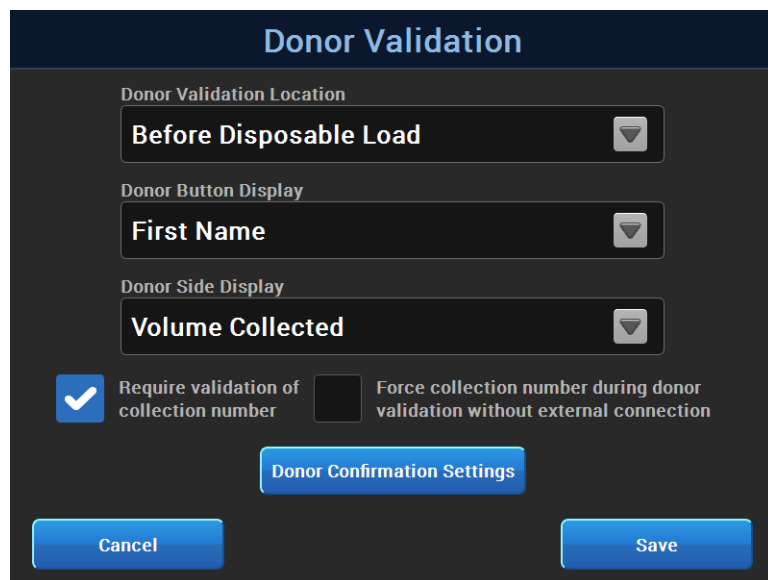


The **Cuff Pressure Settings** screen is divided into two main sections: **DRAW PRESSURE** and **PRESSURE INTERVAL**. The **DRAW PRESSURE** section contains three columns labeled **START**, **MIN**, and **MAX**. Each column has a blue up arrow button above a numerical value and a blue down arrow button below it. The values are 50 for START, 10 for MIN, and 100 for MAX. The **PRESSURE INTERVAL** section has a single column with a blue up arrow button above the value 5 and a blue down arrow button below it. At the bottom of the screen are two blue buttons: **Cancel** on the left and **Save** on the right.

Figure 10, Example of the Cuff Pressure Settings screen

Donor Validation

Touch **Donor Validation** to access the *Donor Validation Settings* screen, which controls various settings that pertain to the donor.



The **Donor Validation** screen features three drop-down menus: **Donor Validation Location** (set to **Before Disposable Load**), **Donor Button Display** (set to **First Name**), and **Donor Side Display** (set to **Volume Collected**). Below these menus are two checkboxes: **Require validation of collection number** (checked) and **Force collection number during donor validation without external connection** (unchecked). A blue button labeled **Donor Confirmation Settings** is positioned between the checkboxes. At the bottom are two blue buttons: **Cancel** on the left and **Save** on the right.

Figure 11, Example of the Donor Validation Settings screen

Donor Validation Location

Touch the Donor Validation Location drop-down menu to select where in the disposable loading sequence the *Donor Information* screen will appear. The *Donor Information* screen displays the donor ID and other information about the donor or procedure and is used to verify the donor.

Donor Button Display

Select what information is displayed on the **Donor** button. The **Donor** button appears in the top banner and provides access to donor information.

Donor Side Display

Select what information appears on the donor side displays. Two donor side displays are located on each side of the device and can be used to communicate information on the collection progress to the donor.

Require Validation of Collection Number

If the device is connected to a compatible software system (or BECS), it should receive a unique collection number for each procedure. If the device is connected but for some reason does not receive a collection number, the device prompts the user to enter a collection number following the *Donor Information* screen. To require the server application to verify the donor collection number that has been entered by the user before the device can proceed to the next screen, select the checkbox on the left. This feature is only available if the device is connected to a compatible software system.

Force Collection Number

Select the checkbox on the right to prompt the user to enter a collection number for the donor if the device is not connected to a compatible software application (e.g., BECS).

Donor Confirmation Settings

Touch **Donor Confirmation Settings** to select which donor information fields will be highlighted on the *Donor Information* screen.

The screenshot shows a 'Donor Confirmation' screen with a table of settings. The table has two columns: 'Donor Information Field' and 'Highlight'. The rows are 'Name', 'Collection Number', 'Plasma Target', and 'Height'. The 'Name' and 'Plasma Target' rows have a checked checkbox in the 'Highlight' column. The 'Collection Number' and 'Height' rows have an unchecked checkbox. There are also up and down arrow buttons next to the 'Collection Number' and 'Height' rows. At the bottom, there is a checkbox labeled 'Require display of selected fields' and a 'Close' button.

Donor Information Field	Highlight
Name	☒
Collection Number	☐
Plasma Target	☒
Height	☐

☐ Require display of selected fields

Close

Figure 12, Example of the Donor Confirmation screen

If the “Require display of selected fields” checkbox is selected, the **Confirm** button on the *Donor Information* screen (See [Figure 13](#)) will not appear until the user touches each highlighted **Show Info** button to view the associated donor information.



Note: To protect the donor’s privacy, the personal information on the Donor Information screen is hidden by default.

Donor Name		Collection #		
Show Info		Show Info		
Plasma Target	Height	Weight	Hematocrit	
Show Info	Show Info	Show Info	Show Info	
Government ID		Date of Birth		Donor ID
Show Info		Show Info		Show Info

Reset Confirm

Figure 13, Example of the Donor Information screen

Saline Options

Touch **Saline Options** to access the saline return options for when a saline protocol is selected. Under “Saline Type,” select a checkbox to set whether saline is automatically dispensed halfway through the procedure, at the end of every cycle, or at the end of the procedure. Under “Saline Parameters,” select the checkbox to enable the user to modify the saline options during a procedure when a saline protocol is selected.

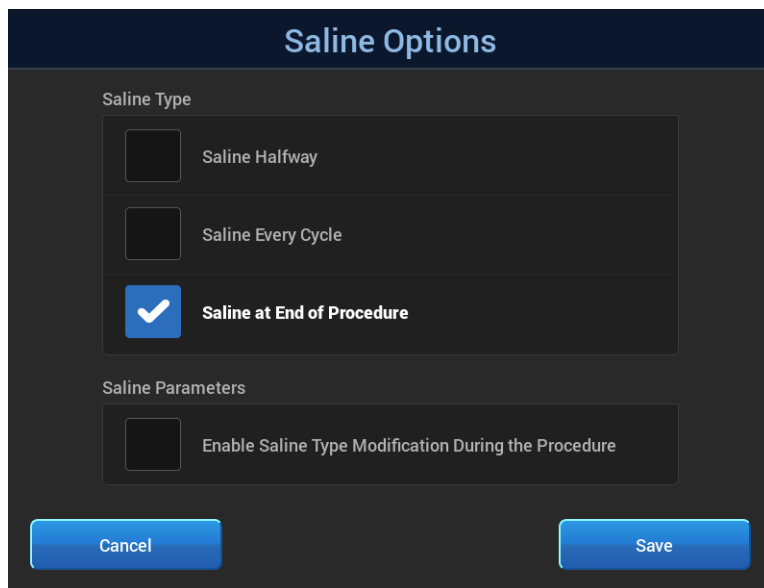
The "Saline Options" screen has a dark blue header with the title "Saline Options" in white. Below the header, there are two sections. The first section, "Saline Type", contains three radio button options: "Saline Halfway", "Saline Every Cycle", and "Saline at End of Procedure". The "Saline at End of Procedure" option is selected, indicated by a blue checkmark in its radio button. The second section, "Saline Parameters", contains a single checkbox option labeled "Enable Saline Type Modification During the Procedure", which is currently unchecked. At the bottom of the screen, there are two blue buttons: "Cancel" on the left and "Save" on the right.

Figure 14, Example of the Saline Options screen

Plasma Targets

Touch **Plasma Targets** to access the list of target collection volumes that appear on the *Donor Information* screen for procedures not connected to a BECS. For connected procedures, the device receives the plasma target from the BECS, so this field will not display on the *Donor Information* screen.

The "Plasma Targets" screen has a dark blue header with the title "Plasma Targets" in white. Below the header, there is a "Plasma Target" section. This section contains a text input field followed by the unit "ml", an "Add" button, a dropdown menu, and a "Remove" button. At the bottom of the screen, there are two blue buttons: "Cancel" on the left and "Save" on the right.

Figure 15, Plasma Targets screen

To add a new target collection volume:

1. Touch the "Plasma Target" field.
2. Type a number.

3. Touch **Add**. The new volume appears in drop-down menu.

To delete a target collection volume:

1. Touch the drop-down menu.
2. Select a volume.
3. Touch **Remove**.



Note: If there are no preconfigured plasma targets and the device is not connected to a BECS, the user will need to manually enter or scan the plasma target (and hematocrit, if configured) on the Donor Information screen.

Barcode Formats

In order for the user to scan the plasma target or hematocrit, a barcode must be generated using one of the following formats.



Note: The device's operating parameters are set to milliliters by default. If the units on your device have been configured to use grams instead, the device will assign "grams" to the scanned plasma target value.



Note: If the barcode does not comply with these formats or the values contained in the barcode are outside the allowed range, the device notifies the user that the barcode is invalid. If the device has preconfigured plasma targets, the plasma target contained in the barcode must match one of the preconfigured values.



Note: Both the built-in barcode reader and the Haemonetics approved Datalogic® Gryphon™ external scanner support standard 1-D and 2-D barcodes.

YES® Standard and YES® Technology Procedures

For standard procedures or when YES Technology is enabled, the barcode must use the following formats:

- To scan only the plasma target, the barcode must consist of three to five digits. Leading zeros are allowed. The following are examples:
 - 00690 (plasma target = 690 mL)
 - 0800 (plasma target = 800 mL)
- To scan only the hematocrit, the barcode must consist of exactly two digits. The following are examples:
 - 40 = 40%
 - 55 = 55%
- To scan both the plasma target and the hematocrit, the barcode must consist of exactly six digits, with the first four digits indicating the plasma target, followed by two digits for the hematocrit. The following are examples:

- 062540 (plasma target = 625; hematocrit = 40%)
- 080055 (plasma target = 800; hematocrit = 55%)

Persona™ Technology Procedures

When Persona Technology is enabled, the barcode must use the format:

HCT#HGT#WGT#

Where HCT, HGT, and WGT stand for hematocrit, height, and weight respectively, and # represents the values for each. For example, the device interprets a barcode formatted as HCT45HGT72WGT174 as:

- Hematocrit = 45%
- Height = 72 in.
- Weight = 174 lb

User Settings

Touch **User Settings** to access the default user authentication settings. Every device is programmed at the manufacturer with a default user ID and password, which enable you to access the device's advanced settings. Haemonetics recommends that you change these credentials upon initial login.



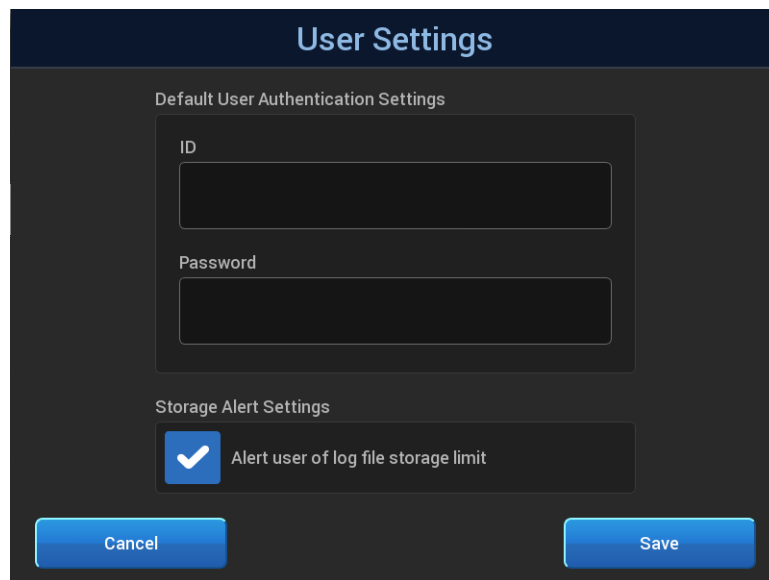
Note: If the device is connected to a BECS, the BECS performs the user authentication.

To change the default user ID and password, select each field and enter a new value.

To alert the user when the device reaches its log file storage limit, ensure the "Storage Alert Settings" checkbox is selected.



Note: Log files will not be automatically deleted or overwritten. Log files can be cleared in the Maintenance screens. (See the Service Manual for reference.)



User Settings

Default User Authentication Settings

ID

Password

Storage Alert Settings

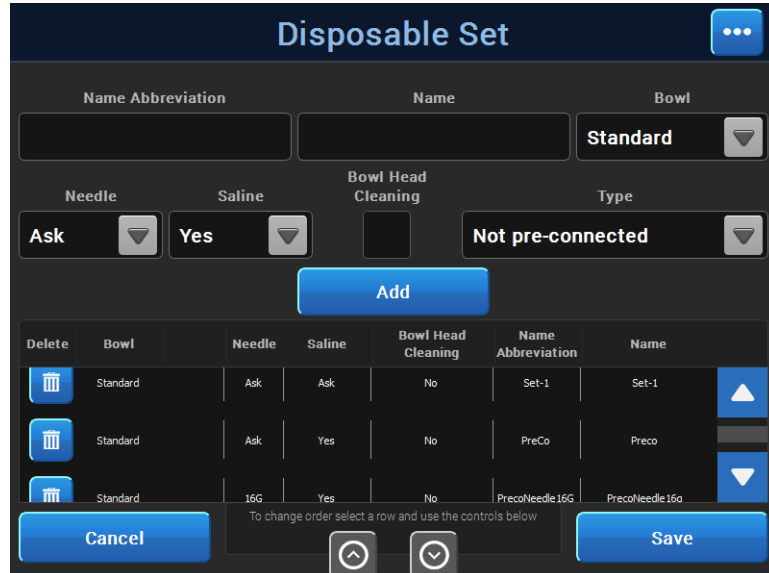
☒ Alert user of log file storage limit

Cancel Save

Figure 16, Example of the User Settings screen

Disposable Set

Touch **Disposable Set** to access the list of disposable sets the user can choose from when beginning a procedure.



Disposable Set

Name Abbreviation Name Bowl

Needle Saline Bowl Head Cleaning Type

Ask Yes Not pre-connected

Add

Delete	Bowl	Needle	Saline	Bowl Head Cleaning	Name Abbreviation	Name
	Standard	Ask	Ask	No	Set-1	Set-1
	Standard	Ask	Yes	No	PreCo	PreCo
	Standard	16G	Yes	No	PreCoNeedle 16G	PreCoNeedle 16G

Cancel Save

To change order select a row and use the controls below

Figure 17, Example of the Disposable Set Settings screen

Adding Disposable Sets

To add a new disposable set:

1. Enter the required information in the top fields.

- a. **Name Abbreviation:** Enter an abbreviated disposable set name. This name will be displayed to the user at disposable set selection during the procedure.
- b. **Name:** Enter a name for the disposable set. This name is communicated to the BECS.
- c. **Bowl:** Select the bowl type.
- d. **Needle:** Select the needle gauge or select “Ask” to prompt the user to select the gauge before advancing to the *Arm Selection* screen.
- e. **Saline:** Select an option for returning saline to the donor.
 - “Yes” - The device returns saline at the end of every cycle.
 - “Ask” - The device prompts the user at the beginning of the procedure to select a return option.
 - “No” - The device does not return any saline to the donor.
- f. **Bowl Head Cleaning:** If a saline option is selected, during every Return cycle the device pumps 10 mL of saline through the bowl header to flush out any residual red blood cells so they are not pushed into the plasma collection container during the Draw cycle. Bowl head cleaning is automatically disabled if the saline option is disabled for the disposable set.
- g. **Type:** Select a type to define whether the disposable set is pre-connected. The options are:
 - Not pre-connected
 - Pre-connected
 - Pre-connected with needle

2. Touch **Add**.

Editing a Disposable Set

To edit a disposable set:

1. Select the set from the list
2. Edit the information that appears in the top fields.
3. Touch **Update**.

Use the arrows at the bottom of the screen to change the order of the sets as they appear in the drop-down menu on the *Disposables Selection* screen.

To remove a disposable set from the list, touch  (**Delete**).

Scan Settings

For additional settings concerning scanning disposable sets, touch  (**More Options**). The screen displays several optional requirements for when the device is connected (“with BECS”) or not connected (“Without BECS”) to a

software system. Select the corresponding checkboxes under each heading to enable or disable any of the following options:



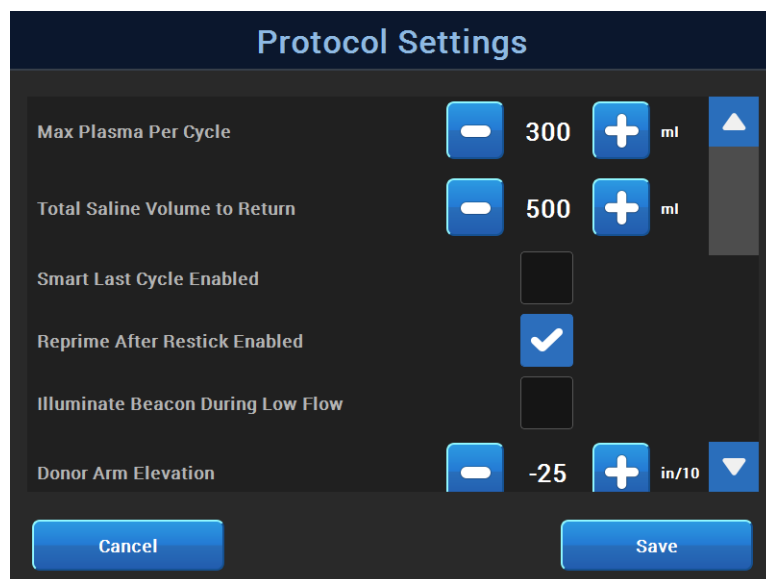
Note: The user may scan the barcodes or enter the information manually.

- **Scan Disposable:** Requires the user to enter all disposables information before proceeding to load the pumps.
- **Scan Needle:** Requires the user to enter the needle information before proceeding to the *Phlebotomy* screen.
- **Scan Needle Sample Only:** Requires the user to enter the needle information for Sample Only procedures before proceeding to the *Sample Only* screen.
- **Read lot number from GS1 bar codes:** Enables the device to read just the lot number for any scanned item that has a valid GS1 barcode. Otherwise, the device reads all of the information contained in the barcode, such as expiration date or GTIN.

Figure 18, Disposable set scanning options

Protocol Settings

Touch **Protocol Settings** to access the protocol settings for each procedure.



The image shows a 'Protocol Settings' screen with a dark background and blue text and buttons. The settings are as follows:

Setting	Value	Unit	Control
Max Plasma Per Cycle	300	ml	Buttons: -, +
Total Saline Volume to Return	500	ml	Buttons: -, +
Smart Last Cycle Enabled	☐		Checkbox
Reprime After Restick Enabled	☒		Checkbox
Illuminate Beacon During Low Flow	☐		Checkbox
Donor Arm Elevation	-25	in/10	Buttons: -, + and a dropdown arrow

At the bottom of the screen are two buttons: 'Cancel' and 'Save'.

Figure 19, Example of the Protocol Settings screen

Table 1, Modifiable Protocol Settings

Parameter	Values	Default
Max Plasma Per Cycle	100 to 300 mL	300 mL
Total Saline Volume to Return	100 to 500 mL	500 mL
Smart Last Cycle Enabled	Checked/Unchecked	Checked
Reprime After Restick Enabled	Checked/Unchecked	Checked
Illuminate Beacon During Low Flow	Checked/Unchecked	Unchecked
Donor Arm Elevation	-120 to 120 in./10	-25 in./10
Empty AC Bag Weight	0 to 500 g	50 g
AC Solution Volume	100 to 1000 mL	250 mL
Low Flow Pump Speed Target Percentage	1 to 99%	75%
Standardized Cycle Mode	Checked/Unchecked	Unchecked
No Arm Selection When Not Connected to BECS	Checked/Unchecked	Unchecked
Plasma Weigher Image	Bottle Holder/Weigher Arm	Bottle Holder
Overdraw Threshold	0 to 30 mL or g	0 mL or g
Underdraw Threshold	0 to 30 mL or g	0 mL or g
Recovery Maximum Delay	0 to 10 min	10 min
Allow Showing Volume in Plasma Load Cell	Checked/Unchecked	Unchecked
Bowl Optics Enabled	Checked/Unchecked	Checked
Maximum Number of Return Flow Errors Allowed Per Cycle	0 to 20	0
Plasma Suction Compensation Offset	0 to 2	0
AC Ratio x 10000	625 to 1250	625
PPP Line Sensor Threshold	50 to 95%	80%
Plasma Density x 1000	1000 to 1050 (g/mL)*1000	1026 (g/mL)*1000
AC Density x 1000	1000 to 1024 (g/mL)*1000	1024 (g/mL)*1000

Table 1, Modifiable Protocol Settings

Parameter	Values	Default
Flow Control Draw Resistance - 16G	0 to 500 mmHg/mL/ min/100	63 mmHg/mL/ min/100
Flow Control Return Resistance - 16G	0 to 500 mmHg/mL/ min/100	148 mmHg/ mL/min/100
Flow Control Saline Resistance - 16G	0 to 500 mmHg/mL/ min/100	26 mmHg/mL/ min/100
Flow Control Draw Resistance - 17G	0 to 500 mmHg/mL/ min/100	80 mmHg/mL/ min/100
Flow Control Return Resistance - 17G	0 to 500 mmHg/mL/ min/100	182 mmHg/ mL/min/100
Flow Control Saline Resistance - 17G	0 to 500 mmHg/mL/ min/100	32 mmHg/mL/ min/100
Skip Load Cell Test	Checked/Unchecked	Unchecked
Plasma Weight Percentage	80 to 100%	100%

Parameters

- **Max Plasma Per Cycle:** Sets the maximum volume of plasma that can be collected per cycle.
- **Total Saline Volume to Return:** Sets the total volume of saline to be returned to the donor over the course of a donation, if a saline protocol is selected.



Note: The device can be programmed to reinfuse saline for volume replacement at the end of each Return cycle, during the final Return cycle, or proportioned halfway through the procedure and then during the final Return cycle with up to 500 mL of replacement fluid.

- **Smart Last Cycle Enabled:** Enables or disables Smart Return. When enabled, the device shortens the Return cycle prior to the final Draw cycle by returning only a portion of the bowl contents to the donor, depending on the volume of plasma remaining to be collected.
- **Reprime After Restick Enabled:** Determines whether the device provides the option to reprime the donor line with blood after a restick to push any air to the blood filter.
- **Illuminate Beacon During Low Flow:** Enables or disables the status beacon for low flow notifications. If enabled, the beacon light glows orange when a low flow notification appears.

- **Donor Arm Elevation:** Sets where the donor's arm is located in relation to the donor pressure monitor. See "[Donor Arm Elevation](#)" for more information.
- **Empty AC Bag Weight:** Sets how much an empty anticoagulant bag weighs.
- **AC Solution Volume:** Sets the nominal volume of the anticoagulant solution bag, which the device uses with the empty AC bag weight to verify the weight of the AC bag during installation.
- **Low Flow Pump Speed Target Percentage:** If the pump speed falls below this percentage of the target pump speed, the device displays the "Low Flow" notification.
- **Standardized Cycle Mode:** If enabled, the device calculates the number of additional cycles needed to reach the plasma target based on the plasma volume collected in the first cycle. It then averages the remaining collection volume over the number of additional cycles with the goal of ending those cycles before detecting cellular fluid at the line sensor.



Note: Standardized Cycle Mode cannot be used with the Smart Last Cycle parameter. If both are enabled, only Standardized Cycle Mode will be implemented.

- **No Arm Selection When Not Connected to BECS:** If enabled, the *Arm Selection* screen is not displayed if the device is not connected to a BECS.
- **Plasma Weigher Image:** Selects the weigher image type displayed on the *Weigher Test* screen.
- **Overdraw Threshold:** If set greater than 0, the operator will be informed if the collected volume exceeds the target volume by this value.
- **Underdraw Threshold:** If set greater than 0, the operator will be informed if the collected volume is lower than the target volume by this value.
- **Recovery Maximum Delay:** Sets the maximum elapsed time allowed for procedure recovery after the device is restarted due to a power failure, safety system fault detection, or other system error.
- **Allow Showing Volume in Plasma Load Cell Test:** If enabled, the *Plasma Weigher Test* screen provides an option to display the weight on the plasma weigher as a volume in milliliters.
- **Bowl Optics Enabled:** Determines whether the bowl optics are used. If disabled, the device operates in Line Sensor Only mode.



Attention: *Use of the 625HS-00 high-separation bowl is prohibited when bowl optics are disabled.*

- **Maximum Number of Return Flow Errors Allowed Per Cycle:** If the number of “High Pressure in Return” errors detected in a single Return cycle is greater than this value, a “Pause” error will be triggered.
- **Plasma Suction Compensation Offset:** If greater than 0, an adjustment is added to the plasma target to compensate for the extra fluid in the effluent tubing that is pulled back into the bowl at the beginning of the Return cycle. For example, if set to 1g, the device stops the final Draw cycle when the plasma weight is 1g more than the target.
- **AC Ratio x 10000:** Sets the AC ratio multiplied by 10000. For example, a ratio of 1/16 (equivalent to 0.0625) is entered as 625.
- **PPP Line Sensor Threshold:** Sets the sensitivity of the line sensor during the later stages of the Draw cycle. Higher values make the line sensor more sensitive to changes, which may cause the device to end the Draw cycle earlier.
- **Plasma Density x 1000:** Sets the plasma density in g/mL with a factor of 1000. For example, the default setting of 1026 represents 1.026 g/mL.
- **AC Density x 1000:** Sets the AC density in g/mL with a factor of 1000. For example, the default setting of 1024 represents 1.024 g/mL.
- **Flow Control Draw/Return/Saline Resistance - 16G/17G:** These parameters are used in the calculation of the estimated donor vein pressure. Slight modification may help with certain disposable sets.
- **Skip Load Cell Test:** Determines whether the plasma weigher test must be performed after powering on the device.
- **Plasma Weight Percentage:** If the collected plasma weight is greater than or equal to this percentage of the target weight at the end of a Draw cycle, the device will initiate a final Return.

Donor Arm Elevation

The “Donor Arm Elevation” setting should be configured as close as possible to the center's bed configurations, within approximately 4 to 6 inches. If centers have donor beds with varying heights, the configuration should be set from the top of the bed height range.

Using a ruler, measure the distance from the top blood filter bracket to the donor bed where the arm will be placed. Then subtract 2.5 inches to account for the average donor arm diameter. Use [Figure 20](#) and the example below for reference.



Note: The setting is in 1/10th of an inch in the software. For example, 2 inches will be set as 20, or 2.5 inches will be 25.

1. Top blood filter bracket
2. Bed arm rest

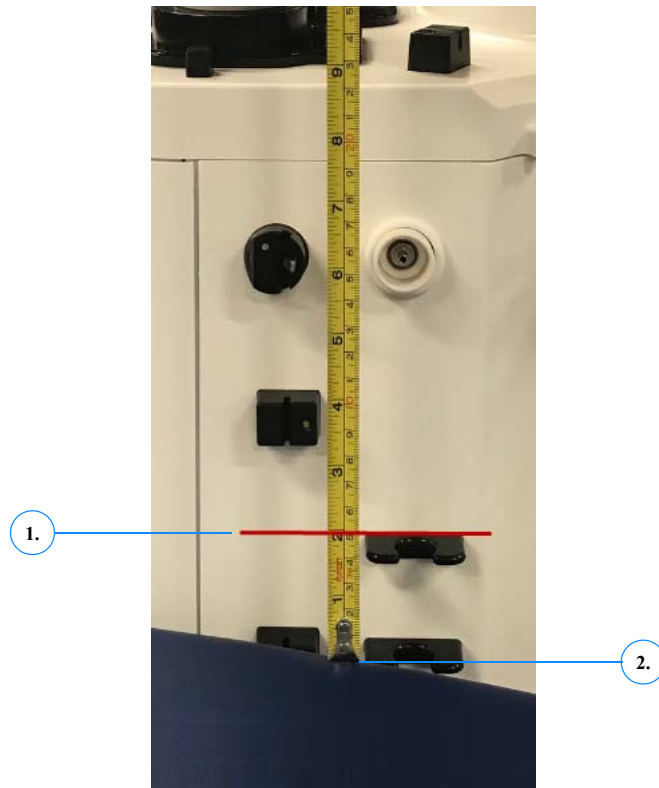


Figure 20, Donor arm elevation from DPM

Donor Elevation (in inches) = arm rest to filter holder height + average arm vertical diameter

Example using measurements in [Figure 20](#):

- Avg. arm diameter = 2.5"
- Distance from bed arm rest to top blood filter bracket = -2"

i Note: Use a negative value when the top of the arm rest is below the top blood filter bracket. If the arm rest is above the blood filter bracket, use a positive value.

- Calculation: $-2" + 2.5" = 0.5"$

The device reads in tenths of an inch. So for this example, you would enter a value of 5 on the device.

Exceptions

Touch **Exceptions** to access the *Exception Handling* screen, where you can configure the exceptions that the user can log during a procedure. An exception is an issue that occurs during a collection procedure that is recorded in the procedure record. Exceptions can be generated automatically, based on configuration settings (For example: an overdraw), or they may be entered manually (For example: a donor reaction).

The screenshot shows a dark-themed interface titled "Exceptions". At the top, there is a text input field labeled "Exception" with two blue buttons to its right: "Add Major" and "Add Minor". Below this, the screen is split into two columns. The left column is titled "Major Exception" and contains a dropdown menu showing "Hypotensive Event" with a downward arrow, and a blue "Remove Major" button below it. The right column is titled "Minor Exception" and contains a dropdown menu showing "Prefaint, No LOC (Minor)" with a downward arrow, and a blue "Remove Minor" button below it. At the bottom of the screen, there are three checkboxes with labels: "Force exception when aborting procedure", "Force exception in case of overdraw or underdraw", and "Allow saving multiple exceptions at once". All three checkboxes are currently unchecked. At the very bottom, there are two blue buttons: "Cancel" on the left and "Save" on the right.

Figure 21, Example of the Exception Handling screen

To add an exception:

1. Touch the "Exception" field. A keyboard appears.
2. Enter a name for the exception.
3. Touch **Add Major** or **Add Minor**, depending on the severity of the exception. It appears as a new entry in the corresponding drop-down menu.

To delete an exception:

1. Select the exception from the appropriate drop-down menu.
2. Touch **Remove Major** or **Remove Minor**, depending on the selection.

Select the "Force exception when aborting procedure" checkbox to require the user to log an exception before aborting a procedure.

Select the "Force exception in case of overdraw or underdraw" checkbox to require the user to log an exception in the event of an overdraw or underdraw before ending a procedure.

Select the "Allow saving multiple exceptions at once" checkbox to enable the user to log more than one exception at a time. This option is disabled by default.

Operator Validation

Touch **Operator Validation** to access the settings for the *Operator Validation* screen, which requires the user to enter a user ID and password before starting or completing a task during a procedure. Select the appropriate checkbox(es) for each point in the procedure when you want the device to prompt the user to log in.

The following conditions control the location of the *Operation Validation* screen in relation to the selected screen or action:

- **Before:** The user must log in before entering the specified screen or performing the specified action.
- **After:** The user must log in after a task has been completed, before exiting the specified state or screen.
- **Confirm:** Requires another user to confirm the previous user's work.

You can require operator validation prompts for the following screens and actions:

- Centrifuge Lock Test
- Weigher Test
- Disposable Set Selection
- Load Disposables
- Load Pumps
- Connect Solutions
- Prime
- Donor Validation
- Arm Selection
- Phlebotomy
- Procedure
- Procedure Complete
- End Procedure
- Adjust Needle
- Reperform Phlebotomy
- Error
- Add Exception
- Sample Only Procedure
- Ignore Checklists
- Maintenance Required
- Change Pump Speed

If you select the “User ID” or “Password” checkboxes at the bottom of the screen the device will display a numeric keyboard instead of a full keyboard for each of those prompts.

Before	After	Confirm
☐ Centrifuge Lock Test	☐	☐
☐ Weigher Test	☐	☐
☐ Disposable Set Selection	☐	☐

Use numeric keyboard for:

☒ User ID

☐ Password

Cancel Save

Figure 22, Example of the Operator Validation screen

Timers

Touch **Timers** to access the disposables, solutions, and end of procedure timers. To enable a timer, select the checkbox next to it. To change the time limit, select the "Hours" or "Minutes" field and enter a new value.

- **Disposables Loaded:** If enabled, the device displays a timer when the disposables are loaded and notifies the user and ends the procedure if the timer ends before entering Draw.
- **Solutions Attached:** If enabled, the device displays a timer upon entering Prime and notifies the user and ends the procedure if the timer ends before entering Draw.
- **End of Procedure:** If enabled, the device displays a timer in the Procedure Complete screen that counts up until the user touches **End Procedure**.

The **Timers** screen displays three active timers, each with a blue checkmark icon in a box. The timers are:

	Hours	Minutes
Disposables Loaded	6	00
Solutions Attached	5	00
End of Procedure		

At the bottom of the screen are two buttons: **Cancel** and **Save**.

Figure 23, Example of the Timers screen

Localization

Touch **Localization** to access the regional parameters, such as:

- Time format
- Date format
- Use of metric units
- Language

The **Localization** screen is divided into three sections:

- Date and Time Format:**
 - Time:** A dropdown menu showing **24 hours**.
 - Date:** A dropdown menu showing **DD/MM/YYYY**.
- Language:** A dropdown menu showing **English**. Below it is a blue button labeled **Install languages**.
- Unit System:** A checkbox labeled **Use metric units**, which is currently unchecked.

At the bottom of the screen are two buttons: **Cancel** and **Save**.

Figure 24, Example of the Localization screen

Installing Additional Languages

English is available by default. Other supported languages must be installed separately. To install a language other than English follow the steps below:



Note: If necessary, contact Haemonetics Product Support for supported languages or to request a language file.

1. Insert the Haemonetics Technician USB into your computer.
2. Create a folder named “translations” (all lowercase letters) at the root directory of the USB flash drive.
3. Unzip the .zip file and place the language file (file extension .tr) into the “translations” folder.



Note: Do not rename the .tr file. It must keep the same name it had in the .zip file.

4. Eject the USB flash drive from the computer and insert it into one of the USB ports on the back of the NexSys PCS device’s touchscreen display.
5. Power on the device and navigate to the *Localization* screen.
6. Touch **Install Languages**. The device installs all language files in the “translations” folder.
7. Once installation is complete, the device displays a “Languages successfully installed” pop-up message. Touch **Confirm**.



*Note: To change the current language, select a language from the drop-down menu and touch **Save**. You must restart the device before the change takes effect.*



Note: If you update the software on a device, you must also reinstall updated language files for non-English languages.

Network Configuration Settings

You can configure your device to connect to a network by accessing the *Network Configurations* screen and either entering the settings manually or scanning a network configuration QR code. Contact your IT Department for the proper settings for your network.



Note: It is the customer's responsibility to ensure the security and integrity of its networks and systems, that its communication networks are configured to support encryption, and that those networks and systems are properly enabled, maintained, and utilized.



Note: In the event of complete loss of the communication network (For example, the router fails, the wireless or Ethernet card on the server fails, there is severe wireless interference with the network, etc.), a disaster recovery plan is recommended. This may include backup hardware and/or a backup paper system.



Note: Various information for the device, including IP address, MAC address, and more, can be found on the System Information screen. See the Service Manual for instructions on how to access this screen.

Data Transfer Options

The NexSys PCS device is capable of both wireless and Ethernet connections. These connections enable the device to exchange donor, device, and procedure information with a software system (also called a BECS) via an approved server application (for example, HaemoCommunicator). Haemonetics highly recommends using a wired Ethernet connection.

Depending on the device's connectivity status, the procedure workflow and screens will vary. During a connected procedure, the device routinely sends and receives information. The NexSys PCS device sends the following information to the approved server application:

- **Status updates:** Messages regarding the current device state
- **Error and notification messages:** Any error or notification messages that may occur to the server application.
- **Technical Data:** The device sends device operational data at the end of each procedure.
- **Disposables information:** Data scanned from barcodes printed on disposables
- **User information:** User validation information from when an operator is required to log in or confirm a previous user's work by entering a user ID and password
- **Procedure records:** The device sends procedure record information to the server application at the end of each procedure. The device must

receive an acknowledgment from the server application before it marks the data as sent.

In exchange, the NexSys PCS device receives the following information from the approved server application:

- **User authorization:** Information relating to if the operator is authenticated and authorized to access the device
- **Collection order:** Information related to the donor collection such as donor demographics, photo, and target collection amount
- **Configuration:** Device configuration settings. You must restart the device before any changes will take effect.

Cybersecurity

It is the customer's responsibility to ensure the security and integrity of its networks and systems, that its communication networks are configured to support encryption, and that those networks and systems are properly enabled, maintained, and utilized.

The NexSys PCS device data that is transferred is encrypted using TLSv1.2 encryption. Additionally, the device allows you to configure the wireless connection to utilize WPA2 security protocol.

Wired Connectivity

The NexSys PCS device supports both wired Ethernet and wireless connectivity. Haemonetics highly recommends using a wired Ethernet connection to communicate with the server application. The Ethernet port is located on the rear of the touch screen display.

Wireless Connectivity

The NexSys PCS device is equipped with an internal wireless antenna and wireless module communication component that enable the device to communicate with an approved server application. Wireless technology provides the following benefits:

- Enables the device to transmit procedure data electronically to an 802.11 g/n compatible wireless access point connected to the local area network (LAN).
- Transmits data in an encrypted standard format (WPA2) that may be used for electronic data collection and management by a central monitoring software system.
- Allows for easier movement of devices by eliminating the need for cabling.
- Serves as a backup in the event that the wired connection fails for any reason.

Wireless Network Setup and Maintenance

If the network uses wireless technology, it is the customer's responsibility to ensure that the wireless network has enough capacity to simultaneously accommodate a connection to all wireless devices required on the network. You should mitigate risks of wireless interference according to the center's needs and SOPs.

Monitor the NexSys PCS device and other devices for symptoms of interference. Interference can be caused by any equipment (Examples: microwaves, refrigerators, Bluetooth® devices, fluorescent lights, etc.) that operates on the same frequency band. Interference may cause delays, timeouts, or lack of access. Relocate equipment, change channels, or follow wireless equipment manufacturer instructions to reduce or eliminate interference.

NexSys PCS devices should be kept a minimum distance away from each other (approximately 39 in. [1 m]) to prevent interference of data transmission. Also, other electronic devices located in the center should be kept a minimum distance away from the NexSys PCS device. To calculate this distance, refer to the IEC/EN 60601-1-2 standard requirements located in the user manual.

The performance of the network depends upon proper setup, primarily selection of the channel, SSID (network name), and unique IP address of each wireless module. Evaluate the network signal level in the room for possible conflicting wireless networks and other RF devices in the vicinity. Check the signal level periodically and whenever there is an installation of new wireless networks or other RF devices.

Creating a Network Configuration QR Code

To generate a network configuration QR code:

1. Using a computer, download the PCSNetConfig utility from the Haemonetics Software Download Center.
 - a. Go to <http://www.haemonetics.com>, select **LOGIN**, and click on **Software Download Center** in the drop-down menu.
 - b. Enter your customer login credentials.



Note: If you do not already have a login, contact Haemonetics Product Support to request access.

- c. Under "HaemoCommunicator," click on **NexSys PCS® Driver**.
 - d. Download the "_dist.zip" file and extract the contents.
2. Navigate to the "pcs300-setup-utility" folder.
3. Open **PCS300NetConfig.htm** and ensure that javascript is enabled.
4. Complete the form with the appropriate network settings and click **Generate QR Code**. The QR code appears.

Device Support Utilities
PCS300 Network Configuration v0.2

Network Configuration		Wireless Configuration	
DHCP:	Enable ▾	Network SSID:	
Static IP Address:		Encryption Type:	WPA2 ▾
Default Gateway:		Authentication Key:	
Subnet Mask:			
Enable/Disable DNS:	Enable ▾	Server Application Configuration	
Primary DNS:		Server Address:	?
Secondary DNS:		TCP Port:	4005
		Site ID:	
		Security Certificate:	? Browse... No file selected.

Figure 25, Example of the network configuration QR code generator

5. Print the QR code and follow the instructions in “Disposable Set” on page 20 to use it to configure the device to connect to an approved network.



Note: For assistance, contact Haemonetics Product Support by calling (800) 356-3506 or by emailing productsupport@haemonetics.com.

